Sustainability

CIVE475 Summer 2025 Life Cycle Assessment

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Objectives of this unit

- Understand the concept of sustainability and its complexity
- Explore the challenges in defining a sustainable product (process or project)
- Be able to describe the role of life cycle assessment within a sustainability framework

Sustainable Development

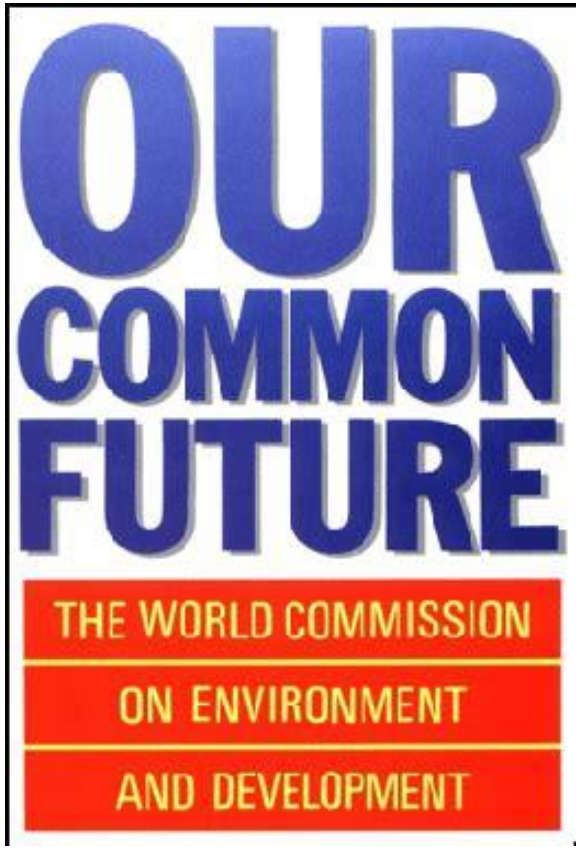
Sustainable Development

Relevant literature: Our common future - Brundtland Report (1987)

> *"Sustainable development is development that meets the needs of the present world without compromising the ability of future generations to meet their own needs."*

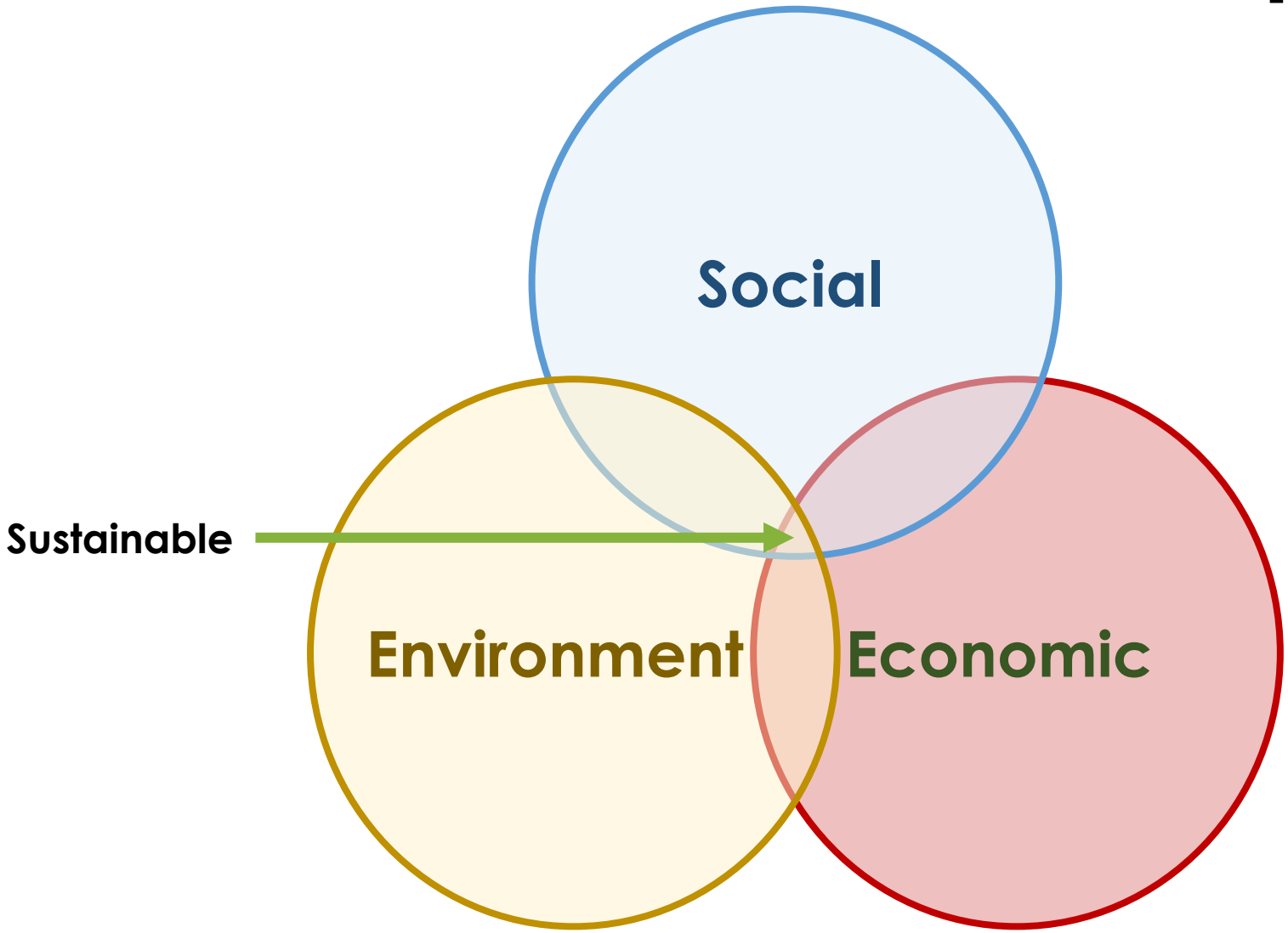
Environmental concern at the time: Chlorofluorocarbon (CFC) and Ozone Layer Depletion

Outcome: Montreal Protocol (1987); Precautionary principle

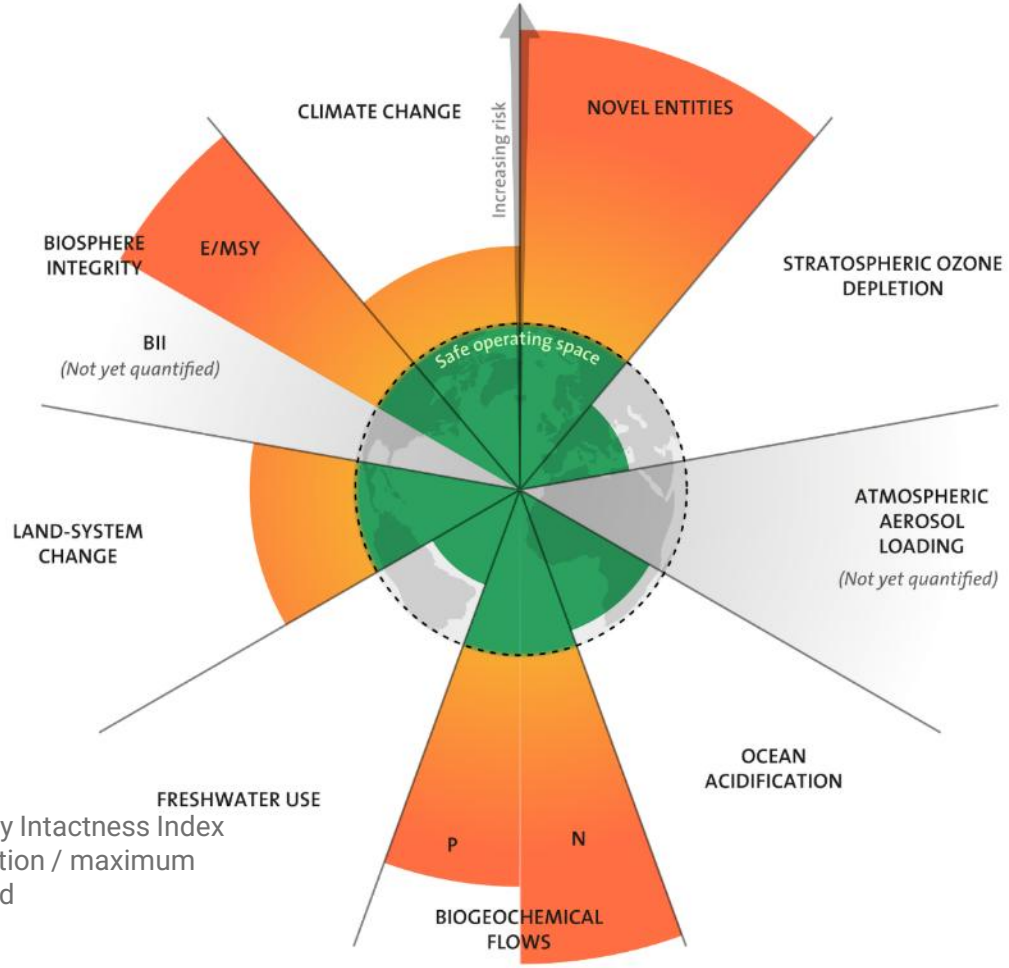


Report of the World Commission on Environment and Development: Our Common Future

3 pillars of Sustainable Development



Planetary Boundaries



BII = Biodiversity Intactness Index
E/MSY = Extinction / maximum sustainable yield

Credit: Designed by Azote for Stockholm Resilience Centre, based on analysis in Persson et al 2022 and Steffen et al 2015

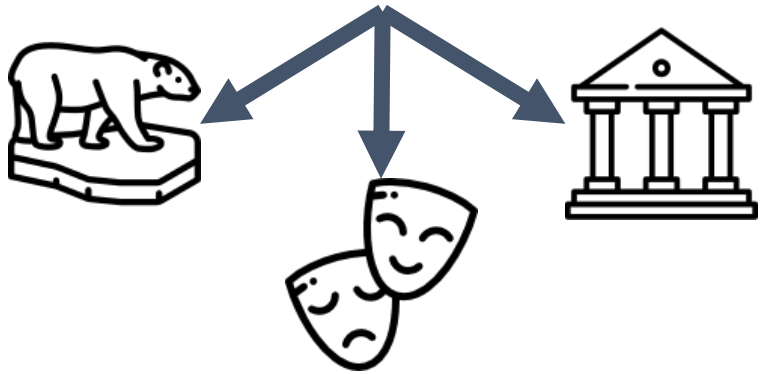
"Transgressing a boundary increases the risk that human activities could inadvertently drive the Earth System into a much less hospitable state, damaging efforts to reduce poverty and leading to a deterioration of human wellbeing in many parts of the world, including wealthy countries."

W. Steffen Australian National University, Canberra

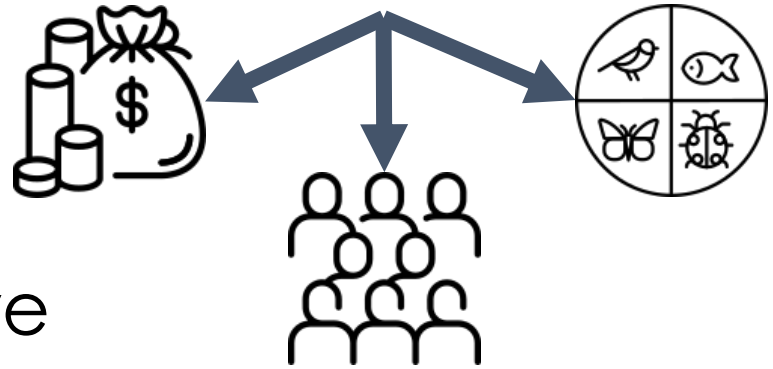
Operationalizing Sustainable Development

What does sustainable development imply?

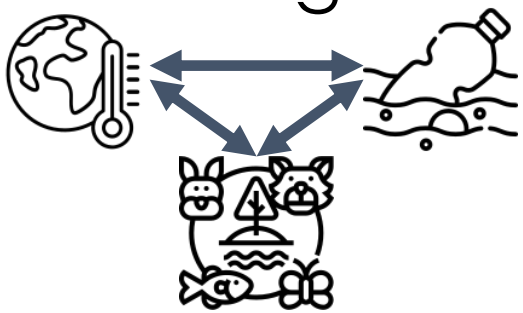
What are we to sustain?



How to measure sustainable development?



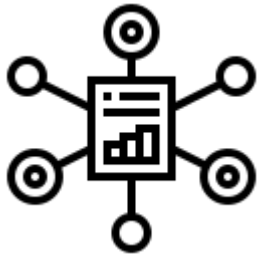
How can we avoid/minimize load shifting?



How to assess sustainable development?

Is it obvious what is sustainable?

- Easy problems have been addressed
- **Sustainable solutions are complex, multidisciplinary, and multifactorial**



Requires a **sustainability assessment framework**, with clear **scope**, **operational methods** and **indicators**

Example of complexity for sustainable solutions

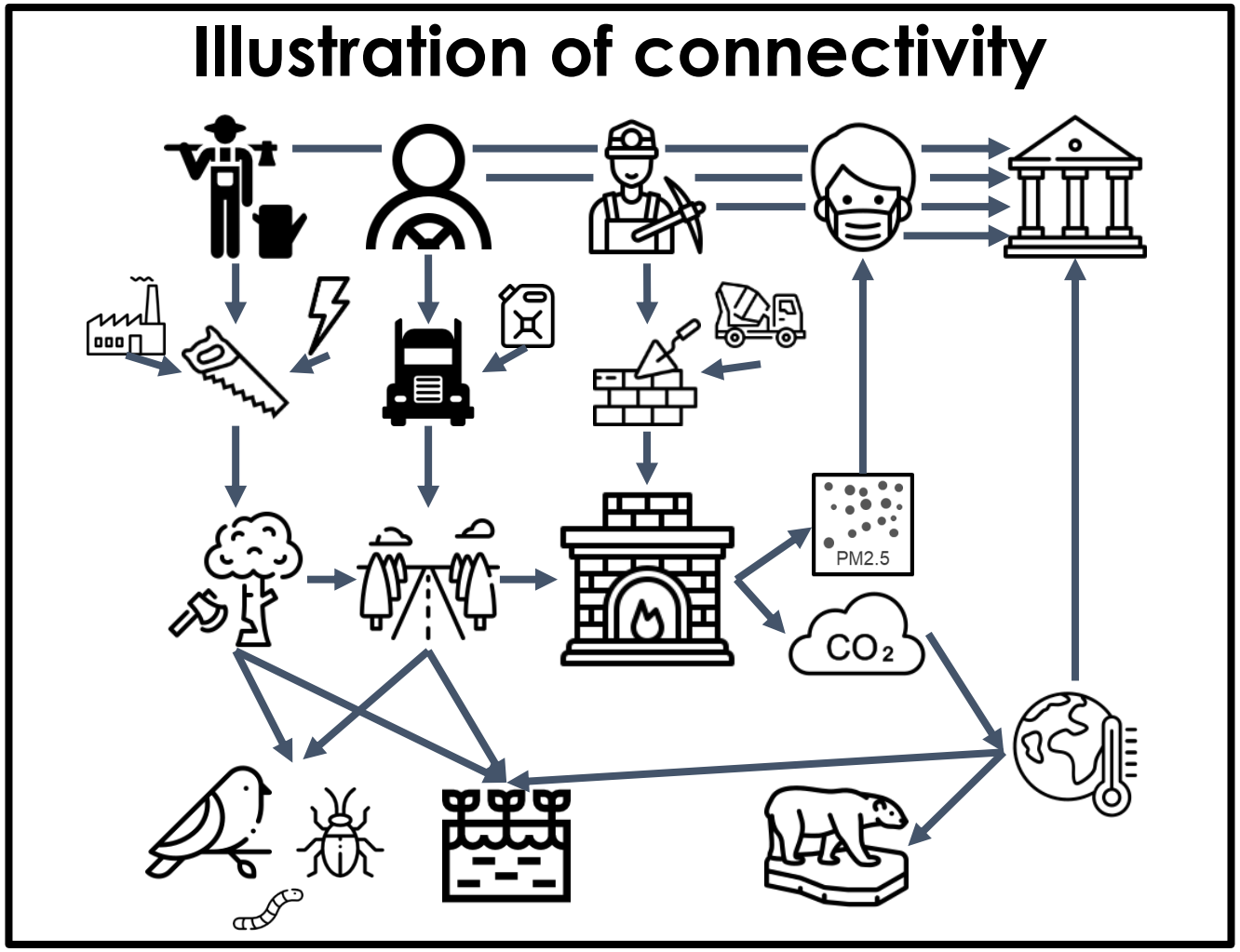
What would be better for the economy, society and environment?

Using trees as energy feedstock

Using trees as construction material

Leaving trees standing

Example of Complexity for Sustainable Solutions



Sustainability assessment framework

Framework: high-level concept that supplies theoretical construct for the assessment

- Brings **meaning** to the analysis, **guides** selection of methods and indicators and **facilitates** interpretation of results
- **No single agreed** upon sustainability assessment framework
- Must be **evidence-based**
- **Further work needed to operationalize integrative sustainability assessment methods**

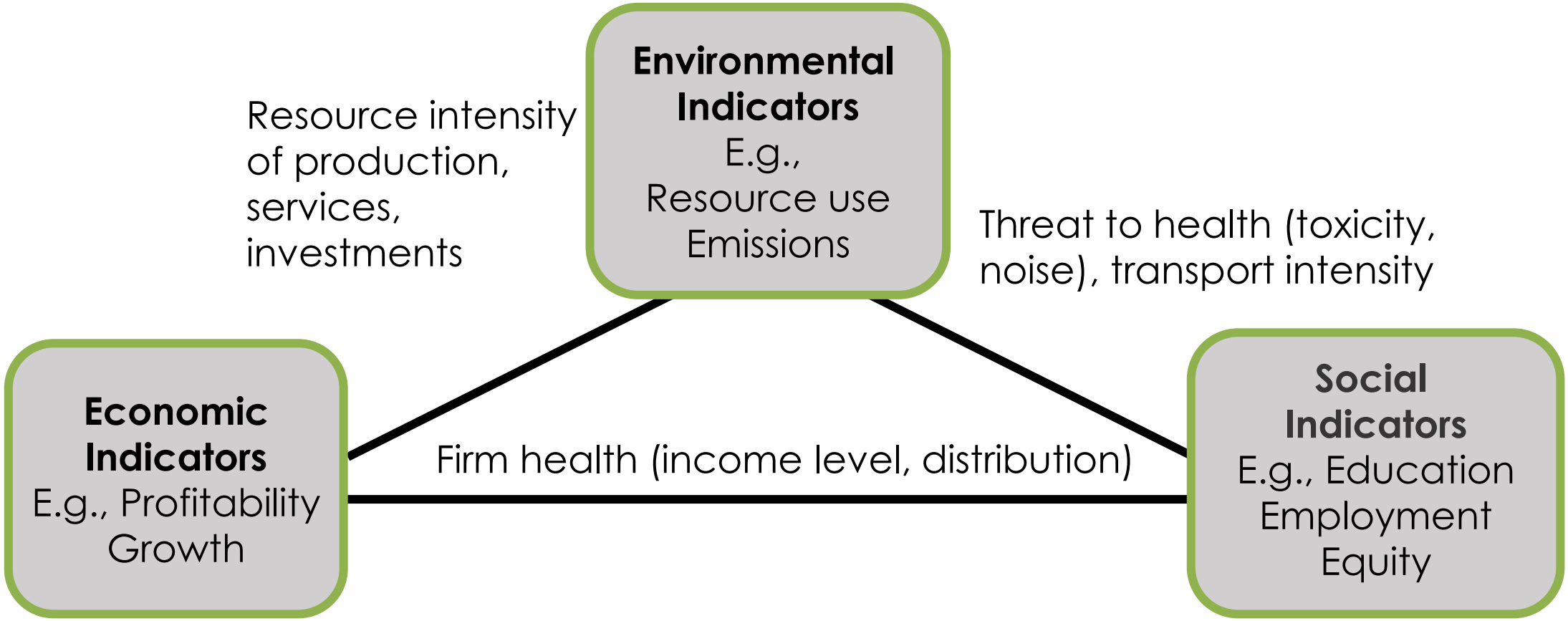
Example: UN Sustainable Development Goals

17 sustainable development goals and 169 targets adopted in 2015 by UN member states

Sustainability indicators

- **Indicators**: Variables that can be measured, derived, estimated or calculated
  - Quantitative estimates of magnitude or severity of an issue
  - Need to fit within the scope of the assessment
  - Need to measure advances toward targets
- **Economic/social indicators** have always been an integral part of governance
  - Environmental dimension was missing until 1990s

Example: Sustainability indicators



System thinking and Life Cycle Thinking

System thinking:

- **Acknowledging** the complex connectivity of our societies, economies and the natural environment
- Understanding **system behaviors** and **responses**

Life cycle thinking:

- Specific example of system thinking applied to **products and services**
- **LCA** is a scientific method that enables life cycle thinking and is widely accepted as the preferred approach to assessing sustainability (UNEP 2009)

Sustainability in Practice

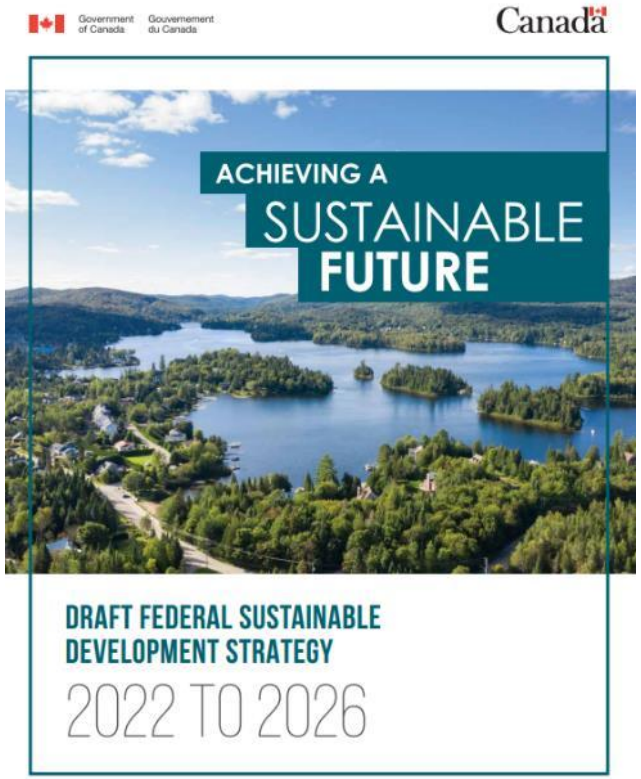
Types of sustainability initiatives

- **International-level** initiatives to track and compare country performance
  - World Bank, United Nations, OECD reporting
- **Individual national efforts** to assess program or agency performance against government targets
- **Corporate initiatives** to develop technologies or market products against sustainability criteria
- **Research initiatives** to further develop and apply methods

Example: Carbon Disclosure Project

- International non-profit organisation
- **Helps investors, companies, cities and governments** measure and act on their environmental impact
- **Develops** a disclosure and scoring system that is constantly evolving
- **Areas of focus**: Climate, water, forests

Example: Sustainable Development in Canada



- **Federal Sustainable Development Strategy** sets out the Government of Canada's environmental sustainability priorities, establishes goals and targets, and identifies actions to achieve them
- **Federal Sustainable Development Act**, passed in Parliament in 2008, provides legal framework for the strategy
- **Example of a target**: reduce Canada's GHG emissions by 40-45% relative to 2005 level by 2030

Example: Corporate initiatives

"Phasing out single-use plastic in all our boxes"

Link

"Carbon negative by 2030, emission removal by 2050"

Link

"100% deforestation-free primary supply chains for meat, palm oil, paper and pulp, soya and sugar"

Link

Example: Circular Economy

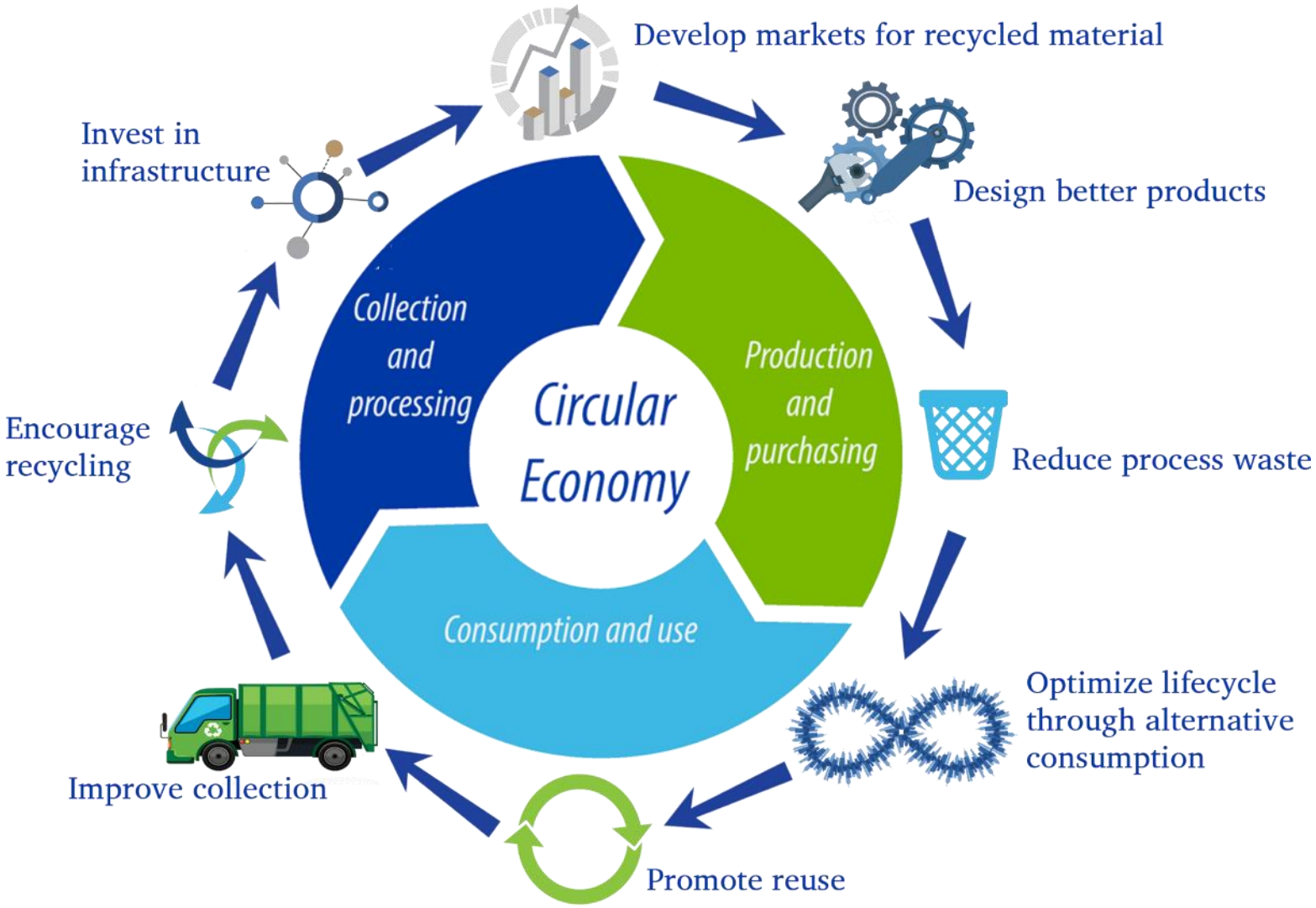


Image: Sustainable Global Resources Ltd.
Recycling Council of Ontario

What have we learned?

- **Environmental, economic and social** performances should be considered in the evaluation of engineering activities
- Incorporating sustainability aspects is simple in the abstract but challenging to operationalize!
  - Need **quantitative** tools and performance metrics that can be applied in engineering design
  - Along with methods for dealing with qualitative aspects
- Role of LCA in sustainability frameworks/assessment